

Microsoft DHCP Options Reference

bootpd Project Documentation | October 2026

1. Microsoft Route Options

Option 249 - Microsoft Classless Static Route (RFC 3442)

Code: 249 | Name: MS-Classless-Static-Route | Data: N (variable)

Format: Each route entry contains:

Destination Bits (N) | Subnet Mask (N) | Gateway (4 bytes)

Example for 10.232.8.0/21 via 10.232.0.1:

0xF9 Option code 249
0x0C Length 12 bytes
0x18 24-bit destination prefix (/21 = 24 bits significant)
0xE8 0x08 10.232 (network portion)
0x00 .0 (network continuation)
0xE8 0x08 0x00 0x01 Gateway 10.232.0.1

Option 121 - Classless Static Route (legacy, pre-Windows XP)

Code: 121 | Name: Classless-Static-Route | Data: N (same format as Option 249)

2. Microsoft Vendor Specific (Option 43)

Option 43 - Vendor Specific Information

Code: 43 | Name: Vendor-Specific-Info | Data: N (variable-length)

Microsoft Option 43 Sub-Options for RRAS/VPN:

Sub-Option 1: RBS Vendor ID = "MSFT 98"
Sub-Option 2: TFTP Timeout = 1 byte (seconds)
Sub-Option 3: RRAS Routing = routing table data
Sub-Option 4: Default Site = site name (ASCII)
Sub-Option 5: Swap Server = WINS server IP
Sub-Option 6: Root Path = NFS/iSCSI root path
Sub-Option 10: Palette Entry = color palette info
Sub-Option 11: Server IP = TFTP server IP

3. Microsoft NAP Options (Network Access Protection)

Option 43 Sub-Option 1: "MSSHIT" = Microsoft NAP status

Option 43 Sub-Option 2: 32-bit flags for NAP state

Used by Windows Vista/7/2008 NAP enforcement.

4. RRAS VPN Options (Routing and Remote Access Service)

Option 249: Pushes static routes to VPN clients

Option 6: DNS servers for VPN network

Option 15: Domain name (e.g. ad.fblipe.de)

Option 44: WINS servers for NetBIOS name resolution

Option 46: NetBIOS Node Type (8 = Hybrid)

5. DHCP Message Types (RFC 4388 Leasequery)

Type 53.1: DHCPDISCOVER	Type 53.9: DHCPLEASEQUERY (RFC 4388)
Type 53.2: DHCPOFFER	Type 53.10: DHCPLEASEUNASSIGNED
Type 53.3: DHCPREQUEST	Type 53.11: DHCPLEASEUNKNOWN
Type 53.4: DHCPDECLINE	Type 53.12: DHCPLEASEACTIVE
Type 53.5: DHCPACK	
Type 53.6: DHCPNAK	
Type 53.7: DHCPRELEASE	
Type 53.8: DHCPINFORM	

6. PXE Boot Options

Option 60: "PXEClient" = Vendor Class Identifier

Option 43: PXE-specific data (see PXE Specification 2.1)

- Sub-Option 6: MTFTP IP address

- Sub-Option 9: MIME Type (optional)

7. Reference Documents

RFC 2132 - DHCP Options and BOOTP Vendor Extensions
RFC 3442 - Classless Static Route Option
RFC 4388 - DHCP Leasequery
RFC 6926 - DHCPv6 Leasequery